

ANDREW LIN

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EDUCATION

University of Waterloo

Bachelor of Computer Science, Co-op, GPA: 3.92

Waterloo, ON

Expected Apr. 2027

- Term Distinction, 2023 – 2025
- Relevant Courses: Data Structures and Algorithms, Logic and Computation, Object-Oriented Software Development

EXPERIENCE

Software Engineer Intern

Jan. 2025 – Apr. 2025

Geotab

Waterloo, ON

- Building data ingestion services and pipelines for 200K commercial vehicles using C#, GCP, Kubernetes, and Terraform
- Spearheaded a data pipeline redesign to support batched vehicle data ingestion, replacing single-message processing with list-based handling, reducing GCP Pub/Sub costs by \$30K annually
- Developed vehicle enrollment services and signal mappings for partnered auto company, integrating data from 10,000 new vehicles into Geotab's platform and increasing subscriptions by 5%

Software Engineer Intern

May 2024 – Aug. 2024

TD Bank

Toronto, ON

- Led development of an Enterprise Innovation team project delivering near real-time translation between clients and bank tellers, built with FastAPI, React, and deployed on GCP Compute Engine and Heroku
- Achieved a 6% increase in translation performance measured by string-based and ML-based metrics (eg. BLEU, COMET) by evaluating and comparing the quality of four different translation models across 38 language pairs
- Integrated auto language detection using Whisper to build a single-device conversation feature

Software Engineer Intern

Sep. 2023 – Dec. 2023

Altis Labs

Toronto, ON

- Built image annotation tools and services for a clinical information platform using TypeScript, GraphQL, and Hasura
- Developed Python scripts to populate databases with 10+ GB of medical data and images for testing, while automating AWS RDS backups for production and development databases using the AWS SDK and Serverless
- Built a comprehensive patient study labeling system by designing database infrastructure and integrating it into the Hasura GraphQL API, using Express.js endpoints for custom input validation on labels' JSON metadata

Quality Assurance Intern

Jan. 2023 – Apr. 2023

Altis Labs

Toronto, ON

- Improved on the GitHub Actions CI/CD pipeline by introducing new frameworks, test reporters, coverage reports, and Python dependency-caching into the project and reduced the runtime of the testing workflow by 68%
- Integrated tox to streamline test execution across 250+ cases, simplifying environment switching with Pipenv
- Wrote over 20 integration and end-to-end tests using Pytest, Jest, and Cypress, automating 30% of manual test cases

PROJECTS

Large Language Model | Python, PyTorch

- Built a Transformer-based large language model with 124M parameters, implementing multi-head self-attention

Live Transcriptions | JavaScript, Chrome API, OpenAI API

- Coded a Chrome extension that generates live transcripts from any web audio source, using the Chrome API and Whisper

Ticket Finder | React, Spring Boot, Java, Maven, MySQL, Docker

- Developed a back-end service and interactive dashboard to keep track of ticket prices, fetching from the Ticketmaster API

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C/C++, Java, C#, SQL, GraphQL, Bash, HTML, CSS

Libraries/Frameworks: React, Node, Express, Spring Boot, .NET, FastAPI, Apollo Client, Serverless, Cypress, PyTorch

Developer Tools: Git, Docker, PostgreSQL, MySQL, AWS, GCP, Terraform, GitHub Actions, Vercel, Hasura, Kubernetes, BigQuery